

Kirlia

Coursework Part II Report

Predictive Modelling and Clustering of Climate Attitudes

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1 Introduction

This report uses the Peoples' Climate Vote 2024 survey United Nations Development Programme (UNDP) and University of Oxford 2024 to examine public support for strengthening national climate commitments. Part II addresses two questions: (i) can support be predicted from demographic and attitudinal information, and do attitudes or demographics matter more; and (ii) how countries can be grouped into a typology based on climate attitudes and how these clusters vary across world regions. We therefore combine supervised classification of support for stronger commitments with unsupervised clustering of country-level attitude profiles.

2 Data and Methods

2.1 Data

The dataset contains nationally representative surveys for many countries, with demographic variables (age, education, gender, country) and climate-attitude questions such as worry, trust, experience of extreme weather and views on climate policies. Survey weights are used to compute weighted means. For the supervised task, the unit of analysis is the Country–Age–Education group and the outcome indicates whether at least 50% of respondents in that group support "Strengthening commitments"; groups labelled "All Ages" or "All Education" are excluded. For clustering we instead use country-level aggregates, retaining rows where Age and Education take their "All" values and excluding the global total. For the supervised sample, the two outcome classes are not perfectly balanced, with more Country–Age–Education groups showing majority support than not, so we report balanced accuracy alongside accuracy and F1 and use class weights to reduce potential bias towards the majority class.

2.2 Predictive modelling

The dependent variable indicates whether a Country–Age–Education group shows majority support for stronger commitments. Let p be the weighted proportion of respondents in that group choosing "Strengthen commitments"; we set $Y = 1$ if $p \geq 0.5$ and $Y = 0$ otherwise.

Predictors fall into two groups: demographic variables (country, age group, education), encoded via one-hot dummies; and attitudinal variables, given by weighted mean responses to items on climate worry, trust, extreme weather experience, perceived impact on major life decisions and support for climate actions. We estimate logistic regression with balanced class weights and a `liblinear` solver, and a random forest classifier with 400 trees, balanced class weights and a small minimum leaf size. For each model we fit a "full" specification including both demographics and attitudes, and an "attitudes-only" specification excluding demographics. An 80/20 stratified train–test split is used. Performance is summarised using accuracy, F1-score and balanced accuracy Powers 2011 on the holdout set, with 5-fold stratified cross-validation as a robustness check. Feature importance is computed using absolute coefficients (logistic regression) or impurity-based importances (random forest) and aggregated over attitudinal and demographic groups.

2.3 Clustering analysis

The clustering analysis is conducted at the country level. For each country we compute weighted mean scores for the attitude items used in the predictive models and the share supporting stronger commitments, so that each row represents an overall national profile. Missing values are imputed using variable-wise medians and features are standardised to have zero mean and unit variance.

We apply k-means clustering for $k \in \{3, 4, 5, 6\}$ and select the number of clusters that maximises the silhouette score. To aid interpretation, we project the standardised country profiles into two

dimensions using Principal Component Analysis (PCA) and plot the resulting clusters. Cluster characteristics are examined by computing mean attitude scores per cluster and visualising the most strongly varying items in a heatmap. Each country is also mapped to a broad world region (Africa, Asia, Europe, North America, South America, Oceania, Americas) to compare regional composition.

3 Results

3.1 Predictive modelling results

Across all four model specifications—full and attitudes-only logistic regression, and full and attitudes-only random forest—out-of-sample performance is extremely high. On the holdout set, accuracy, F1-score and balanced accuracy are all essentially 1.00, with very similar cross-validation scores. Figure 1 shows the F1-scores, which are almost identical across models, indicating that Country–Age–Education groups are highly separable in the selected feature space.

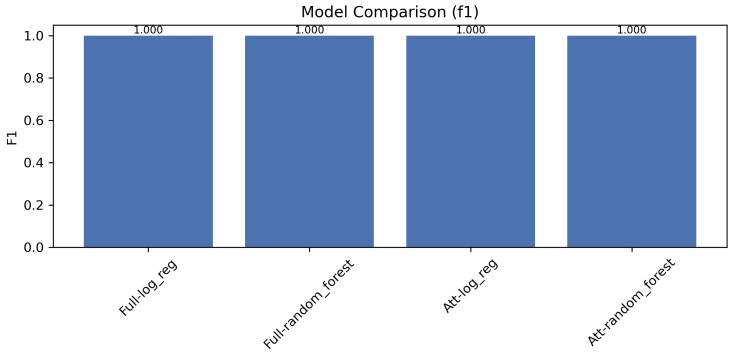


Figure 1. F1-score comparison for full and attitudes-only logistic regression and random forest models (holdout set).

Feature importances from the full random forest model help explain these results. As shown in Figure 2, the ten most important predictors are all attitudinal variables; no demographic feature appears among the top ranks. The strongest effects are associated with climate worry, frequency of thinking about climate change, support for protective actions and evaluations of government and business performance.

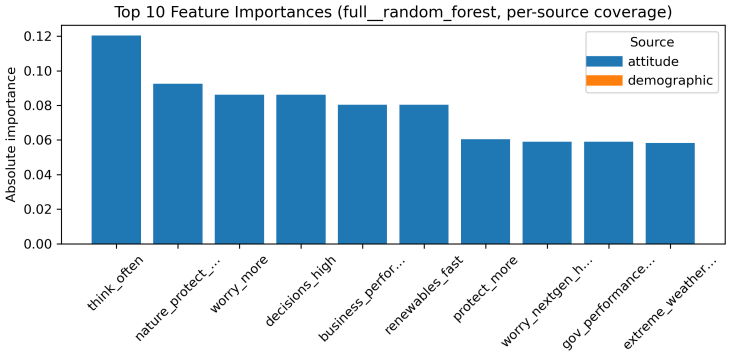


Figure 2. Top ten feature importances for the full random forest model. Highly ranked predictors are all attitudinal variables.

Grouping importances by source confirms that demographics add little. Figure 3 shows that mean absolute importance for demographic predictors is close to zero, whereas attitudinal variables account for virtually all importance. In other words, once attitudes and institutional evaluations are known, demographic information contributes almost no additional predictive power.

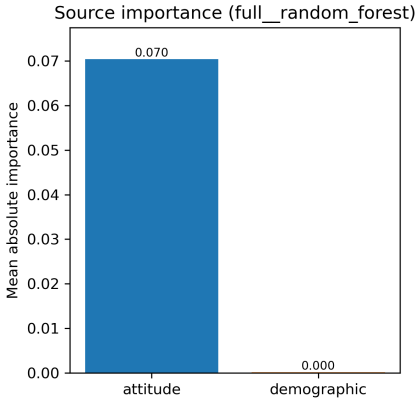


Figure 3. Mean absolute feature importance by source (attitudes vs demographics) in the full random forest model.

3.2 Country clusters

The clustering analysis selects three country clusters, with an average silhouette score of approximately 0.17 and cluster sizes of 32, 11 and 29 countries respectively. Although this value indicates only moderate separation, the PCA projection in Figure 4 shows clear structure and overlap between regions in each cluster, suggesting that national attitude profiles are not purely determined by geography.

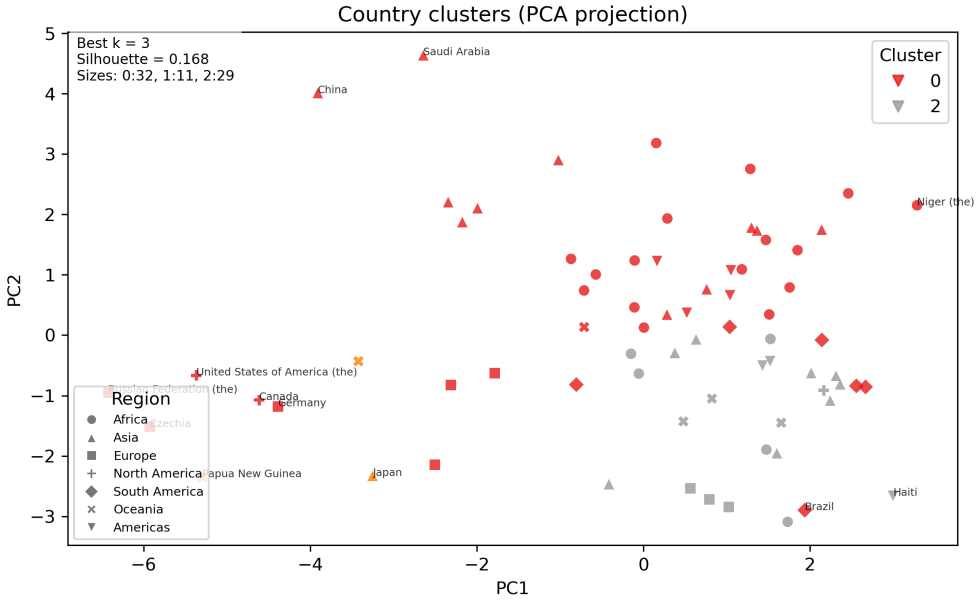


Figure 4. PCA projection of country clusters ($k = 3$). Colours denote clusters; marker shapes denote world regions.

Cluster profiles, summarised in Figure 5, reveal three broad patterns. One cluster is comparatively "less engaged", with lower worry and weaker demand for additional protection, teaching and aid. A second cluster shows high support for protective measures and high worry but more critical views of government and business. The third cluster combines similarly high support with more positive evaluations of institutional performance.

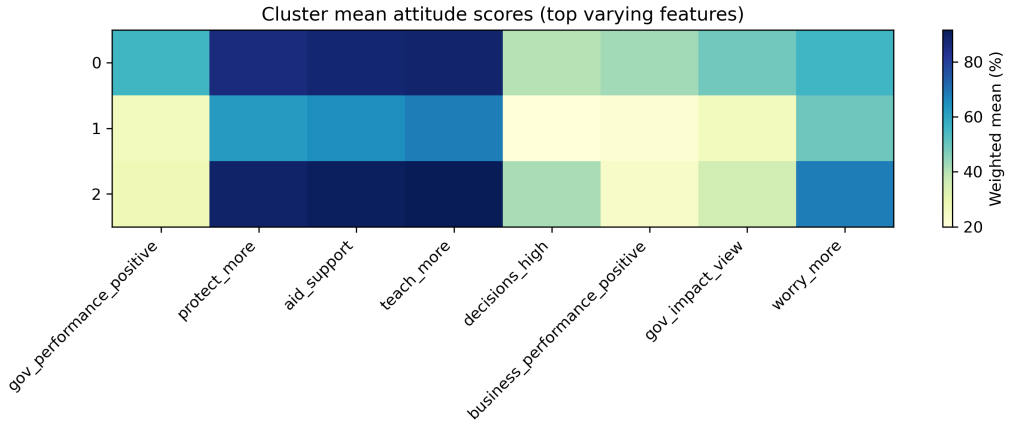


Figure 5. Heatmap of mean attitude scores for the most strongly varying features across clusters. Darker colours indicate higher weighted support.

4 Discussion and Conclusions

Returning to the first research question, the predictive modelling results show that climate-related attitudes and evaluations of institutions almost entirely determine whether a Country–Age–Education group supports strengthening climate commitments. Once variables such as climate worry, perceived risks, extreme weather experience and perceived government and business performance are taken into account, demographic characteristics add virtually no explanatory power. This is consistent with work emphasising the central role of psychological and value-based factors in shaping climate attitudes Leviston, Walker, and Morwinski 2013; Leiserowitz et al. 2021.

With respect to the second research question, the clustering analysis complements these findings by identifying three broad types of countries, ranging from less engaged profiles with weaker support for additional action to highly supportive profiles that differ mainly in trust in institutions. These typologies suggest that communication and policy design should be tailored to national profiles: raising salience and relevance in less engaged contexts, and focusing on institutional credibility where concern is already high. Overall, the analysis highlights that attitudes, rather than demographics alone, are key to understanding and building public support for more ambitious climate commitments.

References

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